DISTRIBUTION OF SHARDS AND REPLICAS

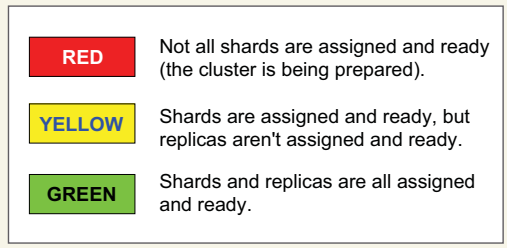
Each shard is expected to hold a certain amount of data. As we know, data is spread across multiple shards on multiple nodes. Let's see how shards are distributed when we start new nodes or diminish when we lose nodes.

Suppose we have created an `virus_mutations` index to hold COVID virus mutation data. According to our strategy, this index will be provided with three shards. (Chapter 6 covers the mechanics of creating an index with a certain number of shards and replicas; for now, let's continue the discussion of how shards are distributed.)

Health status of a cluster

Elasticsearch uses a simple traffic light system based on indicators to let us know a cluster's health at any given time. A cluster has the following three possible states:

- *RED*—The shards have not yet been assigned, so not all data is available for querying. This usually happens when the cluster is starting up, during which time the shards are in a transient state.
- *YELLOW*—Replicas are not yet assigned, but all the shards have been assigned and are in action. This is likely to occur when the nodes hosting the replicas have crashed or are just coming up.
- *GREEN*—This is the happy state when all shards and replicas are assigned and serving as expected.



Indicating the health of shards with a traffic light signal board

Elasticsearch exposes cluster APIs to fetch cluster-related information, including cluster health. We can use a `GET _cluster/health` endpoint to fetch the health indicator; the next figure shows the result of such a call.

```
GET _cluster/health
```

```
{
  "cluster_name" : "elasticsearch",
  "status" : "red",
  "timed_out" : false,
  "number_of_nodes" : 1,
  "number_of_data_nodes" : 1,
  "active_primary_shards" : 25,
  "active_shards" : 25,
  "unassigned_shards" : 24
  ...
}
```

Fetching the status of a cluster by invoking the cluster's health endpoint